On the Strictness of Negative Imaginariness for Linear Systems

Tianzhe Zhang, Chao Chen, Sei Zhen Khong and Alexander Lanzon

***Abstract*— This paper studies the strictness of negative imaginariness for linear systems, with emphasis on the presence of nonzero resonant modes. To accommodate such resonant modes, we introduce the notion of marginally strictly negative imaginary (MSNI) systems, defined through a decomposition into a lossless negative imaginary part and a strictly negative imaginary part. This class permits the nonzero resonant poles needed for resonant controller design and retains the strictness of negative imaginariness away from those poles. Based on this MSNI framework, we establish a DC-gain stability test for the positive feedback interconnection of an MSNI system and a stable negative imaginary system under easily verifiable pointwise assumptions. A numerical example illustrates the validity and effectiveness of the proposed result under resonant tracking requirements.**

I. INTRODUCTION

Negative Imaginary (NI) systems theory was introduced in [1]. It was originally developed as a powerful framework for analysing feedback interconnections that arise in lightly damped mechanical structures and flexible systems with collocated force actuators and position sensors [2]. A notable feature is that for a positive feedback interconnection, closed-loop stability can often be certified through a simple and elegant DC-gain condition. This is attractive because it avoids frequency-by-frequency stability checks and supports a clean design and verification process [1]–[9]. Over nearly two decades, NI systems theory has been extended well beyond this original class of applications in several directions. These directions include lossless NI systems [10]–[12], infinite-dimensional NI systems [13], nonlinear systems [14]–[16], descriptor and state-space characterisations [17], [18], discrete-time NI theory [19], [20], NI systems with free body motion [21], [22], and networked and cooperative control settings [23], [24].

Classical NI stability results for positive feedback interconnections typically require at least one system to be strictly negative imaginary (SNI) [1], [4], [5], [10], [21]. By definition, SNI systems are stable and hence cannot have simple nonzero imaginary-axis poles. However, such poles are deliberately introduced in resonant controllers to achieve tracking or disturbance rejection near prescribed nonzero frequencies [25]–[28]. This naturally leads to the following question. Can we retain a DC-gain stability test while allowing these nonzero resonant poles in the system that provides NI strictness?

Two existing research directions are particularly relevant to this question. The first considers alternative notions of NI strictness, such as output strictly negative imaginary systems and related variants [18], [29]–[31]. These notions extend the available NI strictness formulations in directions different from the classical SNI behaviour, and some of them permit poles at the origin [30]. However, these strictness notions are generally not tailored to accommodate simple imaginary-axis poles away from the origin. Indeed, an undamped resonant mode can sustain a non-decaying oscillation in the output after the input is removed, which conflicts with strict output inequalities built from time-accumulated quadratic terms in the output derivative [29], [30]. Thus, although output strict notions extend NI strictness beyond the classical SNI requirement in some directions, they do not directly resolve the case most relevant here, namely, prescribed nonzero resonant poles on the NI-strictness-providing side.

The second route studies stability when both systems in the positive feedback interconnection are NI. In this setting, each system may contain nonzero imaginary-axis poles, possibly at different locations [32]. This line of work relaxes the use of an SNI system in the feedback interconnection. However, it still requires additional NI strictness together with nonsingularity assumptions to conduct the feedback stability analysis [32]. These requirements increase the verification burden and move the stability test away from a purely DC-gain condition. Across both routes, existing stability results typically treat nonzero imaginary-axis poles and NI strictness separately, with two features assigned to different systems in the feedback interconnection [18], [26], [29], [30], [32]. However, both features may be present in the same system.

To bridge the gap, we introduce the notion of marginally strictly negative imaginary (MSNI) systems, partly inspired by the notions of marginally strictly positive real systems [33] and systems phase [34], [35]. An MSNI system admits a decomposition into a lossless NI part and an SNI part. The lossless NI part carries the nonzero imaginary-axis poles needed for resonant controller design, while the

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SNI part preserves NI strictness away from those poles. This decomposition allows resonant poles to be included on the NI-strictness-providing side without sacrificing the classical DC-gain stability test. It also reduces the additional verification to simple pointwise checks on the stable NI system at the corresponding pole locations. The main contributions of this paper are as follows. First, we introduce the class of MSNI systems and state their basic properties. Second, we develop a DC-gain stability test for the positive feedback interconnection of an MSNI system and a stable NI system.

The remainder of the paper is organised as follows. Section II recalls the NI subclasses and the positive feedback setting used in the paper, and it states the classical NI DC-gain stability test. Section III introduces MSNI systems and states their basic properties. Section IV presents the feedback stability result for the positive feedback interconnection of a stable NI system and an MSNI system. Section V presents an illustrative numerical example. Section VI concludes the paper.

II. Preliminaries

This section introduces the notation and recalls the NI subclasses needed for the MSNI definition. It also recalls the positive feedback setting and the classical stability theorem for the positive feedback interconnection of an NI system and an SNI system.

A. Notation

Let $\mathbb{C}$ denote the complex plane, where $\mathbb{C}_{>0}$ and $\mathbb{C}_{\geq 0}$ denote the open and closed right-half planes respectively. Let $\mathbb{R}^{n\times n}$ and $\mathbb{C}^{n\times n}$ denote the sets of $n \times n$ real and complex matrices respectively. Let $\mathcal{R}^{n\times n}$ denote the set of $n \times n$, real, rational, proper transfer function matrices, and $\mathcal{RH}_\infty^{n\times n}$ denote the subset of stable transfer function matrices in $\mathcal{R}^{n\times n}$, i.e., those with no poles in $\mathbb{C}_{\geq 0}$. The identity matrix of compatible dimension is denoted by I. For a matrix A, the transpose and complex conjugate transpose are denoted by $A^\top$ and A^* respectively. If all eigenvalues of A are real, then $\lambda_{\max}(A)$ denotes the largest eigenvalue of A. A matrix $A \in \mathbb{C}^{n\times n}$ is Hermitian if $A = A^*$, and a matrix $A \in \mathbb{R}^{n\times n}$ is symmetric if $A = A^\top$. For a Hermitian or real symmetric matrix A, $A \geq 0$ (resp. $A > 0$) means A is positive semidefinite (resp. positive definite).

B. Negative Imaginary Systems

Background on NI, SNI, and lossless NI systems can be found in [1], [3], [10]. Only the definitions needed for the MSNI results are recalled here.

For a transfer function matrix G, define the set of its positive imaginary-axis pole frequencies as

$$\Omega_G = \{\omega \in (0,\infty) : s = j\omega \text{ is a pole of } G(s)\}.$$

In this paper, we do not consider poles at the origin.

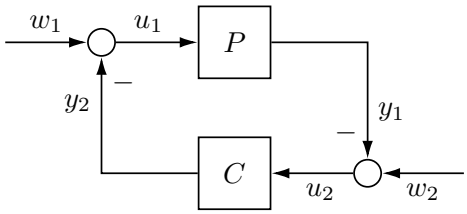


Fig. 1. A positive feedback interconnection of $P, C \in \mathcal{R}^{n\times n}$.

Definition 1 ([3]): A system $G \in \mathcal{R}^{n\times n}$ is said to be negative imaginary (NI) if

1) G has no poles at the origin and in $\mathbb{C}_{>0}$;
2) $j[G(j\omega) - G(j\omega)^*] \geq 0$ for all $\omega \in (0,\infty) \setminus \Omega_G$;
3) for each $\omega_0 \in \Omega_G$, the pole at $s = j\omega_0$ is simple and the residue matrix $\lim_{s\to j\omega_0} (s - j\omega_0)\, jG(s)$ is Hermitian and positive semidefinite.

Lossless NI systems form a special subclass of the NI class, where the inequality in Definition 1 holds with equality for all $\omega \in (0,\infty) \setminus \Omega_G$.

Definition 2 ([10]): A system $G \in \mathcal{R}^{n\times n}$ is said to be lossless negative imaginary (lossless NI) if G is NI and for all $\omega \in (0,\infty) \setminus \Omega_G$,

$$j[G(j\omega) - G(j\omega)^*] = 0.$$

SNI systems form a stable subclass of the NI class and are characterised by a strict frequency domain inequality on the frequency interval $(0,\infty)$.

Definition 3 ([1]): A system $G \in \mathcal{RH}_\infty^{n\times n}$ is said to be strictly negative imaginary (SNI) if for all $\omega \in (0,\infty)$,

$$j[G(j\omega) - G(j\omega)^*] > 0.$$

C. Preliminaries on Positive Feedback Interconnections and NI Stability Results

This subsection fixes the positive feedback interconnection and the stability notion used in the paper. It also states the classical NI DC-gain stability test.

Consider a positive feedback interconnection shown in Fig. 1, where $P \in \mathcal{R}^{n\times n}$ and $C \in \mathcal{R}^{n\times n}$, w_1 and w_2 are external signals, and $u_1,\ u_2,\ y_1$ and y_2 are internal signals. Such a positive feedback interconnection is denoted by $[P, C]$ and is described by

$$\begin{bmatrix} w_1 \\ w_2 \end{bmatrix} = \begin{bmatrix} I & C \\ P & I \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \end{bmatrix}.$$

Definition 4 ([36, Lemma 5.3]): A positive feedback interconnection of P and C is said to be *stable* if the transfer function

$$\begin{bmatrix} I & C \\ P & I \end{bmatrix}^{-1} = \begin{bmatrix} I + C(I-PC)^{-1}P & -C(I-PC)^{-1} \\ -(I-PC)^{-1}P & (I-PC)^{-1} \end{bmatrix}$$

from (w_1, w_2) to (u_1, u_2) belongs to $\mathcal{RH}_\infty^{2n\times 2n}$.

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In [1], [5], the following classical NI stability result is stated for the positive feedback interconnection.

Theorem 1 ([1], [5]): Let $P \in \mathcal{R}^{n\times n}$ be NI and $C \in \mathcal{RH}_\infty^{n\times n}$ be SNI. Suppose that at least one of them is strictly proper. Then, the positive feedback interconnection $[P, C]$ is stable if and only if

$$\lambda_{\max}(P(0)C(0)) < 1. \tag{1}$$

The result yields a clean DC-gain test (1) provided that one of the two systems is SNI. The SNI assumption rules out simple nonzero imaginary-axis poles on that side. We argue that this restriction can be relaxed by introducing the MSNI notion in the next section.

III. MSNI Systems

This section introduces MSNI systems. It begins with a motivating example that highlights the need to accommodate resonant poles and motivates the decomposition used in the MSNI definition.

A. A Motivating Example

In the following SISO example, the system P is NI, and the system C contains an SNI part. Since both P and the SNI part of C are stable, the setting remains aligned with Theorem 1. However, the SNI part in C alone is not sufficient to reflect the resonant control action introduced at a prescribed nonzero frequency. For this reason, we introduce a resonant part in C. This part introduces a pair of simple imaginary-axis poles and shapes the closed-loop behaviour near the selected frequency [26]–[28]. The resulting resonant structure is consistent with the internal model principle and supports tracking or disturbance rejection at that frequency [25].

Example 1: Consider

$$P(s) = \frac{2s^2+s+1}{(s+0.5)(s+1)(s^2+2s+5)},$$
$$C(s) = \frac{1}{s^2+1} + \frac{1}{s+1}.$$

All the poles of P lie in the open left-half plane, so $P \in \mathcal{RH}_\infty$. Moreover, P is NI. The positive-frequency Nyquist plot in Fig. 2 provides a graphical illustration of this property and shows that $P(j)$ lies on the real axis, so the NI inequality is not strict at $\omega = 1$ rad/s. For C, the first term $1/(s^2+1)$ is lossless NI and has simple poles at $s = \pm j$, while the second term $1/(s+1)$ is SNI. Since $P(j) = 0.2$, the poles of $C(s)$ at $s = \pm j$ are not cancelled in $P(s)C(s)$. A numerical root check of $1 - P(s)C(s)$ indicates that all closed-loop poles lie in the open left-half plane. Hence, the positive feedback interconnection is stable by [36, Theorem 5.7].

In this example $P(0)C(0) = 0.8$, and both P and C are strictly proper. Together with $P(j) \neq 0$, this indicates that the DC-gain test may be sufficient when the resonant poles are not cancelled in $P(s)C(s)$.

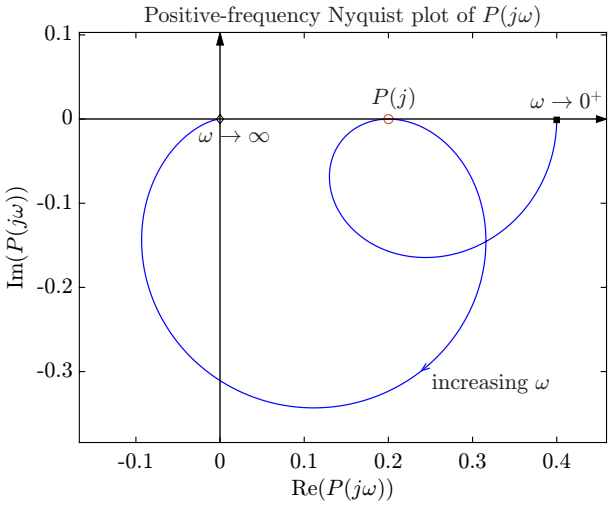


Fig. 2. The Nyquist plot of $P(j\omega)$ over positive frequencies, where $P(s) = \frac{2s^2+s+1}{(s+0.5)(s+1)(s^2+2s+5)}$ is given in Example 1.

B. MSNI Definition

We now introduce the MSNI definition based on the decomposition suggested above.

Definition 5 (MSNI system): A system $G \in \mathcal{R}^{n\times n}$ is said to be *marginally strictly negative imaginary (MSNI)* if it admits the following decomposition

$$G = G_{\mathrm{m}} + G_{\mathrm{s}},$$

where G_{m} is lossless NI, $G_{\mathrm{m}}(\infty) = G(\infty)$, and G_{s} is SNI.

The following remark shows that SNI behaviour is retained away from the imaginary-axis poles.

Remark 1: Since $G_{\mathrm{s}} \in \mathcal{RH}_\infty^{n\times n}$, it has no poles on the imaginary axis. Hence $\Omega_{G_{\mathrm{s}}} = \emptyset$ and $\Omega_G = \Omega_{G_{\mathrm{m}}}$. Since G_{m} is lossless NI, for every $\omega \in (0,\infty) \setminus \Omega_G$,

$$j[G_{\mathrm{m}}(j\omega) - G_{\mathrm{m}}(j\omega)^*] = 0,$$

and therefore

$$j[G(j\omega) - G(j\omega)^*] = j[G_{\mathrm{s}}(j\omega) - G_{\mathrm{s}}(j\omega)^*] > 0.$$

The SNI term G_{s} is essential for enforcing this SNI behaviour away from the resonant poles.

For instance, a simple SISO example is

$$G(s) = \frac{k_{\mathrm{m}}}{s^2+\omega_0^2} + \frac{k_{\mathrm{s}}}{s+a},$$

with $k_{\mathrm{m}} > 0$, $k_{\mathrm{s}} > 0$, $\omega_0 > 0$, and $a > 0$, where the first term is lossless NI and carries the poles at $s = \pm j\omega_0$, and the second term is SNI. Hence G is MSNI by Definition 5. The system C in Example 1 corresponds to the choice $k_{\mathrm{m}} = k_{\mathrm{s}} = 1$, $\omega_0 = 1$, and $a = 1$.

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IV. Feedback Stability Results

This section extends the classical NI stability result recalled in Section II to the case where the SNI system C is allowed to have simple nonzero imaginary-axis poles. The additional issue is to exclude unstable pole–zero cancellation in the closed right-half plane, as mentioned in Example 1. That is, the unstable poles of the loop gain $P(s)C(s)$, counted with multiplicity, should equal the sum of the unstable poles of $P(s)$ and $C(s)$. In the MIMO setting, this is enforced by assumptions of full normal rank [36, Section 3.11] together with the absence of transmission zeros of P at the nonzero imaginary-axis pole locations of C. Under these assumptions, the stability condition can still be stated as the same DC-gain test.

Theorem 2: Let $P \in \mathcal{RH}_\infty^{n\times n}$ be NI with full normal rank and $C \in \mathcal{R}^{n\times n}$ be MSNI. Suppose that at least one of them is strictly proper. Suppose that $P(s)$ has no transmission zeros at any point $s = j\omega$ where $C(s)$ has a pole. Then, the positive feedback interconnection $[P, C]$ is stable if and only if

$$\lambda_{\max}(P(0)C(0)) < 1.$$

Proof: The proof of Theorem 2 is omitted here and can be found in the extended version of this paper [37]. ■

Theorem 2 yields the following observations. In the SISO case, the additional assumption reduces to $P(j\omega) \neq 0$ for every imaginary-axis pole $j\omega$ of C. The stability condition then becomes the scalar test $P(0)C(0) < 1$ together with finitely many pointwise nonzero checks. Example 1 is a direct instance of this SISO case.

Returning to the general MIMO setting, Theorem 2 preserves the classical DC-gain condition in Theorem 1. The additional requirements are that $P \in \mathcal{RH}_\infty^{n\times n}$ have full normal rank and no transmission zeros at the imaginary-axis pole locations of C, which in our square MIMO setting is equivalent to requiring that P be nonsingular at those pole locations [36, Corollary 3.30]. By contrast, the results in [5] assume an SNI system in the positive feedback interconnection and thus rule out imaginary-axis poles on that side. Several related results also require pointwise NI strictness on the system at the imaginary-axis pole locations of the other system [18], [26], [30], [32]. In the present MSNI setting, the NI strictness is already provided by the SNI part of C, so no additional pointwise NI strictness is required on the stable NI side. This point is illustrated in the next section by an instructive numerical example.

V. Numerical Example

This section provides a MIMO example that illustrates the stability result in Section IV and demonstrates resonant tracking on a selected channel.

The positive feedback interconnection is shown in Fig. 1. The closed-loop transfer function matrix from w_2 to y_1 is denoted by

$$T(s) = -(I - P(s)C(s))^{-1}P(s)C(s).$$

We first check the assumptions in Theorem 2 for the chosen plant and controller.

A. Stability Verification

Consider the coupled plant

$$P(s) = Q\begin{bmatrix} \dfrac{1}{s+1} & 0 \\ 0 & 1 \end{bmatrix} Q^\top$$

with

$$Q = \frac{1}{\sqrt{2}}\begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}.$$

Expanding the above expression gives

$$P(s) = \frac{\begin{bmatrix} s+2 & -s \\ -s & s+2 \end{bmatrix}}{2(s+1)}.$$

The controller is chosen as

$$C(s) = C_{\mathrm{m}}(s) + C_{\mathrm{s}}(s)$$

with

$$C_{\mathrm{m}}(s) = \begin{bmatrix} \dfrac{1}{s^2+4} & 0 \\ 0 & 0 \end{bmatrix},$$

$$C_{\mathrm{s}}(s) = \frac{1}{s+2}I.$$

The plant is stable since the only pole of P is at $s = -1$. Moreover, it is NI. For any $\omega > 0$,

$$\begin{aligned} j[P(j\omega) - P(j\omega)^*] &= Q\begin{bmatrix} \dfrac{2\omega}{1+\omega^2} & 0 \\ 0 & 0 \end{bmatrix} Q^\top \\ &= \frac{\omega}{1+\omega^2}\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} \geq 0. \end{aligned}$$

Hence, P satisfies the NI inequality on $(0, \infty)$. Note that the NI inequality is not strict. In fact, $\mathrm{rank}(j[P(j\omega) - P(j\omega)^*]) = 1$ for all $\omega > 0$. Thus $j[P(j\omega) - P(j\omega)^*]$ is rank deficient for every $\omega > 0$. In particular, no pointwise NI strictness is available at the resonant frequency $\omega_0 = 2\ \mathrm{rad/s}$ used in the controller.

For the controller, C_{m} is lossless NI and carries simple poles at $s = \pm 2j$ in the $(1,1)$ entry. Moreover, C_{s} is SNI since $j[C_{\mathrm{s}}(j\omega) - C_{\mathrm{s}}(j\omega)^*] > 0$ for all $\omega > 0$. Therefore, C is MSNI by Definition 5. Since C has simple poles at $s = \pm 2j$, it is not SNI. Hence, Theorem 1 is not directly applicable to this feedback interconnection $[P, C]$.

The controller has imaginary-axis poles only at $s = \pm 2j$, so the additional check in Theorem 2 is required only at $s = 2j$. A direct evaluation gives $\det(P(2j)) = 0.2 - 0.4j$, and hence $P(2j)$ is nonsingular. Moreover, $\lambda_{\max}(P(0)C(0)) = 0.75$. Therefore, the positive feedback interconnection $[P, C]$ is stable by Theorem 2. It is also not directly covered by existing NI stability results, because P does not satisfy

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pointwise NI strictness at the pole location of C [18], [30], [32]. Nevertheless, Theorem 2 still applies and yields the stability conclusion. We next examine the closed-loop behaviour near the resonant frequency.

B. Resonant Tracking

To illustrate resonant tracking on the selected channel, set $w_2(t) = [\sin(\omega_0 t),\, 0]^\top$ with $\omega_0 = 2$ rad/s. The transfer function from $[w_2]_1$ to $[y_1]_1$ is the $(1,1)$ entry of $T(s)$, namely $T_{11}(s)$. As shown in Fig. 3, $|T_{11}(j\omega)| \approx 1$ and $\angle T_{11}(j\omega) \approx 0$ for all ω in a neighbourhood of ω_0. Fig. 4 shows the corresponding response $[y_1(t)]_1$. In steady state, $[y_1(t)]_1$ tracks $[w_2(t)]_1 = \sin(\omega_0 t)$ closely.

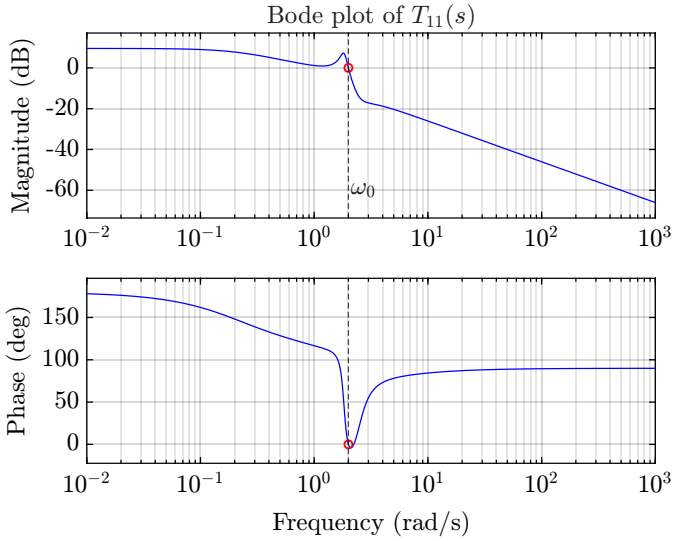


Fig. 3. Bode plot of $T_{11}(s)$, the closed-loop transfer function from $[w_2]_1$ to $[y_1]_1$, with $\omega_0 = 2$ rad/s.

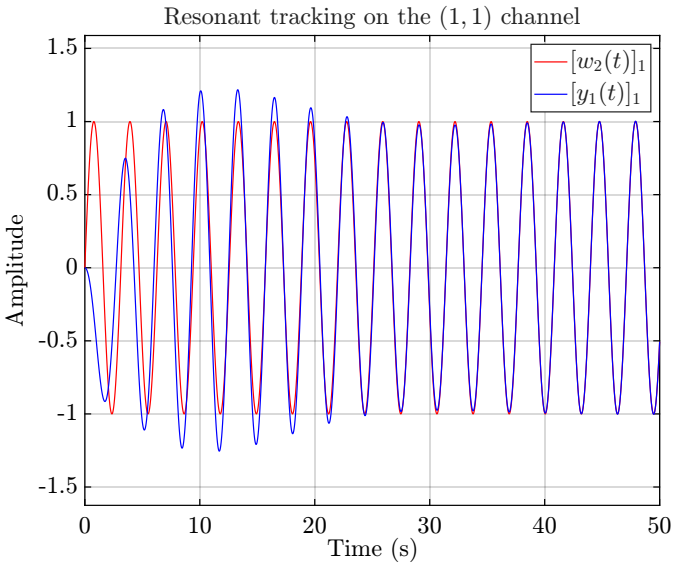


Fig. 4. Time response of $[y_1(t)]_1$ to $[w_2(t)]_1 = \sin(\omega_0 t)$ with $\omega_0 = 2$ rad/s.

VI. Conclusion

This paper studies the role of NI strictness in the presence of nonzero resonant poles. To accommodate such poles within the NI framework, we introduced marginally strictly negative imaginary systems through a decomposition into a lossless NI part and an SNI part. This decomposition makes it possible to place resonant poles on the NI-strictness-providing side. Based on this framework, we established a stability result for the positive feedback interconnection of an MSNI system and a stable NI system. The main point is that the classical DC-gain stability test is retained, with only pointwise nonsingularity checks required at the resonant pole locations. A numerical example was provided to illustrate both the ease of checking the additional assumption and the ability of the proposed framework to support resonant tracking.

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